GOVERNMENT POLYTECHNIC, AMRAVATI

(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)

CURRICULUM DEVELOPMENT CELL

PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING/INFORMATION TECHNOLOGY
COURSE CODE: FC 5463
COURSE TITLE: SOFTWARE TESTING

TEACHING SCHEME:

LEVEL OF COURSE	PRERE-QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
V	-	03	-	02	05	16	48	-	32

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HRS.	ESE		PA	TOTAL	ESE	PA	
3Hrs	MAX.	70	30*	100	25#	25^	150
	MIN.	28	---	40	10	10	

@: Internal Assessment #: External assessment-Practical based $: online examination
(*) Under the Theory PA, Out Of 30 Marks, 20 Marks is the Average of Two Tests and 10 Marks are for Micro project-
(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.
For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

1. RATIONALE:

In today's software environment writing bug-free code is challenging task, which make software testing important tool to get the quality software. Testing techniques include the process of executing a program or application with the intent of finding software bugs and verifying that the software product is fit for use. Students will learn the way to find bugs by applying types, levels and methods of software testing on applications with effective test planning approach. It also covers manual testing.

2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to: -
1) Apply various software testing methods.
2) Prepare test cases for different types and levels of testing.
3) Prepare test plan for an application.
4) Identify bugs to create defect report of given application.
5) Test software for performance measures using automated testing tools.
6) Apply different testing tools for a given application.

3. DETAILED CONTENTS: THEORY

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hrs	Marks
Unit 1. Basics of Software Testing and Testing Methods	1a. Identify errors and bugs in the given program. 1b. Prepare test case for the given application. 1c. Describe the Entry and Exit Criteria for the given test application. 1d. Validate the given application using V model in relation with quality assurance. 1e. Describe features of the given testing method.	1.1 Software Testing, Objectives of Testing. 1.2 Failure, Error, Fault, Defect, Bug Terminology. 1.3 Test Case, When to Start and Stop Testing of Software (Entry and Exit Criteria). 1.4 Verification and Validation (V Model), Quality Assurance, Quality Control. 1.5 Methods of Testing: Static and dynamic Testing 1.6 The box approach: White Box Testing: Inspections, Walkthroughs, Technical Reviews, Functional Testing, Code Coverage Testing, Code Complexity Testing. 1.7 Black Box Testing: Requirement Based Testing, Boundary Value Analysis, Equivalence Partitioning,	**1**	**06**	**08**
Unit 2. Types and Levels of Testing	2a.Apply specified testing level for a web based application. 2b. Apply Acceptance testing for given web based application. 2c. Apply the given performance testing for an application. 2d.Generate test cases for the given application using regression and GUI testing.	2.1 Levels of testing Unit Testing: Driver, Stub Integration Testing: Top-Down Integration, Bottom-Up Integration, Bi-Directional Integration 2.2 Testing on Web Application: Performance Testing: Load Testing, Stress Testing. Security Testing. Client-Server Testing 2.3 Acceptance Testing: Alpha Testing and Beta Testing. Special Tests: Regression Testing, GUI Testing,	**2**	**10**	**14**
Unit 3. Test Management	3a. Prepare test plan for the given application. 3b. Identify the resource requirement of the given application. 3c. Prepare test cases for the given application. 3d. Prepare test report of executed test cases for given application.	3.1 Test Planning : Preparing a Test Plan, Deciding Test Approach, Setting Up Criteria for Testing, Identifying Responsibilities, Staffing, Resource Requirements, Test Deliverables, Testing Tasks 3.2 Test Management: Test Infrastructure Management, Test People Management. 3.3 Test Process: Base Lining a	**3**	**10**	**12**

		Test Plan, Test Case Specification. 3.4 Test Reporting: Executing Test Cases, Preparing Test Summary Report.			
Unit 4. Defect Management	4a. Classify defects on the basis estimated impact. 4b. Prepare defect template on the given application. 4c. Apply defect management process on the given application. 4d.Write procedure to find defect using the given technique.	4.1 Defect Classification, Defect Management Process. 4.2 Defect Life Cycle, Defect Template 4.3 Estimate Expected Impact of a Defect, Techniques for Finding Defects, Reporting a Defect.	**4**	**08**	**12**
Unit 5. Automated Tools and Measurements	5a. Improve testing efficiency using automated tool for given application. 5b. Identify different testing tools to test the given application. 5c. Describe Metrics and Measurement for the given application. 5d. Explain Object oriented metrics used in the given testing application	5.1 Manual Testing and Need for Automated Testing Tools, Advantages and disadvantages of Using Tools 5.2 Selecting a Testing Tool 5.3 When to Use Automated Test Tools, Testing Using Automated Tools. 5.4 Metrics and Measurement: Types of Metrics, Product Metrics and Process Metrics, Object oriented metrics in testing.	**5**	**08**	**12**
Unit 6. Use of different Software testing tools	6a.Able to use bug tracking Tool 6b. Able to create automation script 6c. Apply testing tool for given application	6.1 Selenium Testing Tool: Features, Web driver, Use of selenium to write tests, Methods, 6.2 Mantis Bug Tracker: Features, User interface, Creating Projects, Reporting bugs 6.3 Test link: Features of Test link, Creating Test projects, Creating Test plan, Creating test suite, Creating test case, Assigning test case to test Plan	**6**	**06**	**12**

4. LIST OF PRACTICALS:

Sr No.	PRACTICAL OUTCOMES (PrOs)	CO NO.
1	Identify system specification & design test cases for purchase order Management.	1
2	Analyze and design test cases for simple calculator application.(BB Testing)	1
3	Design test cases for e-commerce (Flipkart, Amazon) login form	2
4	Design test cases for Web Pages Testing any Web Sites	2
5	Prepare test plan for an identified Mobile application.	3
6	Design test plan and test cases for Notepad (MS Window based) Application.	3
7	Prepare defect report after executing test cases for Withdrawn of amount from ATM Machine.	4
8	Prepare defect report after executing test cases for any login form.	4
9	Design and run test cases for WordPad (MS Windows based). Using an Automated tool.	5
10	Design and run test cases for MS Word application using an Automation Tool.	5
11	Test given website (Google, Facebook) using selenium.	6
12	Perform Bug tracking of a given application using mantis Bug Tracker.	6

Note

i. A suggestive list of PrOs is given in the above table. More such PrOs can be added to attain the COs and competency. All above practicals need to be performed so that the student reaches the 'Precision Level' of Dave's 'Psychomotor Domain Taxonomy' as generally required by the industry.

ii. The 'Process' and 'Product' related skills associated with each PrO are to be assessed according to a suggested sample given below:

Sr. No.	Performance Indicators	Weightage in %
1	Preparation of system specification, designing test plan using MS Excel.	40
2	Preparation of defect report	20
3	Execution of test cases using automation tool.	20
4	Answer to sample questions	10
5	Submit report in time	10
	Total	100

The above PrOs also comprise of the following social skills/attitudes which are Affective Domain Outcomes (ADOs) that are best developed through the laboratory/field based experiences:

a. Follow safety practices.
b. Practice good housekeeping.
c. Demonstrate working as a leader/a team member.
d. Maintain tools and equipment.
e. Follow ethical Practices.

The ADOs are not specific to any one PrO, but are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he undertakes a series of practical experiences over a period of time. Moreover, the level of achievement of the ADOs according to Krathwohl's 'Affective Domain Taxonomy' should gradually increase as planned below:

- 'Valuing Level' in 1st year
- 'Organizing Level' in 2nd year
- 'Characterizing Level' in 3rd year

5. SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related co-curricular activities which can be undertaken to accelerate the attainment of the various outcomes in this course.

a. Prepare journal based on practical performed in laboratory.
b. Give seminar on relevant topic.
c. Use different open source software testing for developed softwares.
d. Undertake micro-projects.

6. SUGGESTED INSTRUCTIONAL STRATEGIES

Following are suggested instructional strategies, which the teacher can adopt for the attainment of the various outcomes in this course:

a. Massive open online courses (***MOOCs***) may be used to teach various topics/sub topics.
b. The teacher needs to ensure to create opportunities and provisions for ***co-curricular activities***.
c. Guide student(s) in undertaking micro-projects.
d. Arrange visit to nearby industries and workshops for understanding use of testing tools for testing of softwares.
e. Use automated tool to explain various types of testing.
f. About 10-15% of the topics/sub-topics/contents which is relatively simpler or descriptive in nature may be given to the students for ***self-directed learning*** and assess the development of the COs through classroom presentations. Keep the record of the topics/sub-topics/contents given to the students.

7. SUGGESTED MICRO-PROJECTS.

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. In all semesters, the micro-project are group-based (5-6 students) to build up skill and confidence in every student to become problem solver so that he/she contributes to the projects of the industry. The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than ***16 (sixteen) student engagement hours*** during the course.The student ought to submit micro-project by the end of the semester to develop the industry oriented COs.

A suggestive list of micro-projects is given here. Similar micro-projects could be added by the concerned faculty:

a) Library management :book issue/book stock system
b) E-commerce website testing.
c) Generate test plan for a given application.
d) Generate defect reports for a given application.
e) Develop a test plan a given project.
f) Any other micro-projects suggested by subject faculty on similar line.

8. MAJOR EQUIPMENTS/INSTRUMENTS REQUIRED

Sr No.	Equipment Name with Broad Specification	Practical No.
1	Computer system (Any computer system with basic configuration)	All
2	Selenium	3,11
3	Mantis Bug Tracker	12
4	IBM Rational Functional Tester	5
5	Spreadsheet Package	1,2,3
6	Bugzila	4

9. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Marks per Unit	Distribution of Theory Marks		
			R Level	U Level	A Level
1	Basics of Software Testing and Testing Method	**08**	**04**	**04**	**-**
2	Types and Levels of Testing	**14**	**04**	**06**	**04**
3	Test Management	**12**	**02**	**04**	**06**
4	Defect Management	**12**	**04**	**04**	**04**
5	Testing Tools and Measurements	**12**	**02**	**04**	**06**
6	Use of different Software testing tools	**12**	**02**	**04**	**06**
Total		**70**	**14**	**30**	**26**

R= Remember, **U**= Understanding, **A**=Application and above (*Bloom's Revised taxonomy)*
***Note**: This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of UOs. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table.*

9. SUGGESTED LEARNING RESOURCES:

Sr. No.	Title Of Book	Author	Publication
1.	Software Testing: Principles and Practices	Srinivasan Desikan Gopalaswamy Ramesh	PEARSON Publisher: Pearson India Edition: 1st Edition, 2005 ISBN: 9788177581218, 817758121X
2.	Software Testing: Principles, Techniques and Tools	Limaye M. G.	Tata McGraw Hill Education, New Delhi. Edition: 1st Edition, 2007 ISBN 10: 0070139903 / ISBN 13: 9780070139909
3.	Software Testing: Principles and Practices	Chauhan Naresh	Oxford University Press Noida – 201301, Uttar Pradesh, Edition: 5th Edition
4.	Software Testing	Singh Yogesh	Cambridge University Press, Bangluru. ISBN 978-1-107-65278-1

11. SUGGESTED SOFTWARE/LEARNING WEBSITES

a. http://www.selenium.com
b. http://en.wikipedia.org/wiki/Test_automation
c. http://en.wikipedia.org/wiki/Software_testing#Testing_tools
d. http://www.softwaretestingsoftware.com
e. www.toolsqa.com